

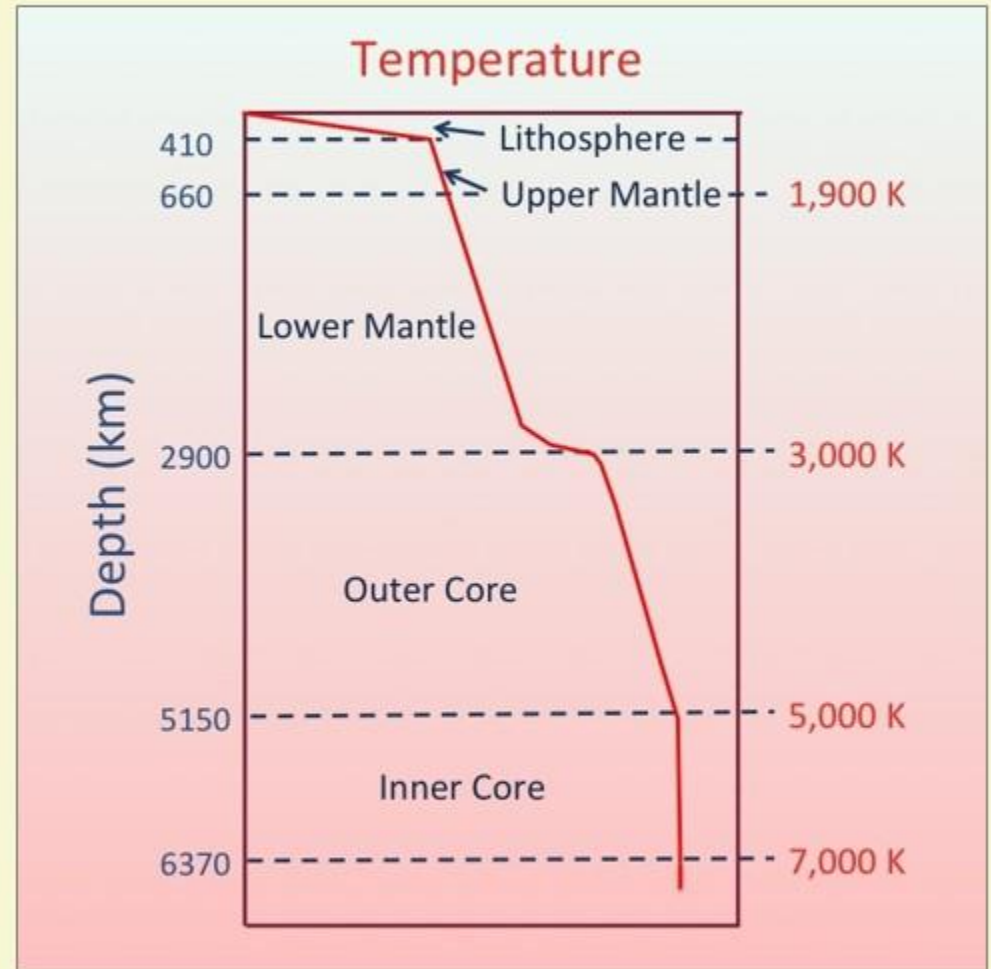
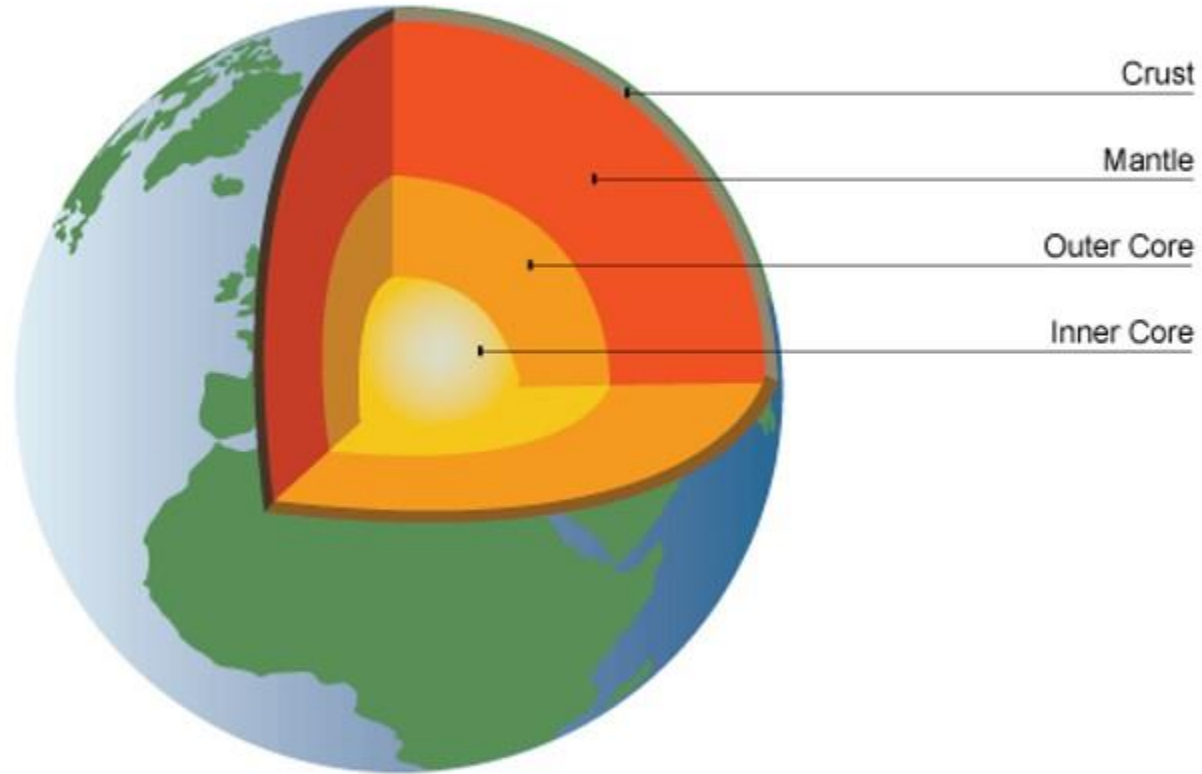
GEOGRAPHY

PHYSICAL GEOGRAPHY



MODULE - 13

- EARTH'S CRUST
- MOHO DISCONTINUITY
- LITHOSPHERE
- TECTONIC PLATES



EARTH'S CRUST

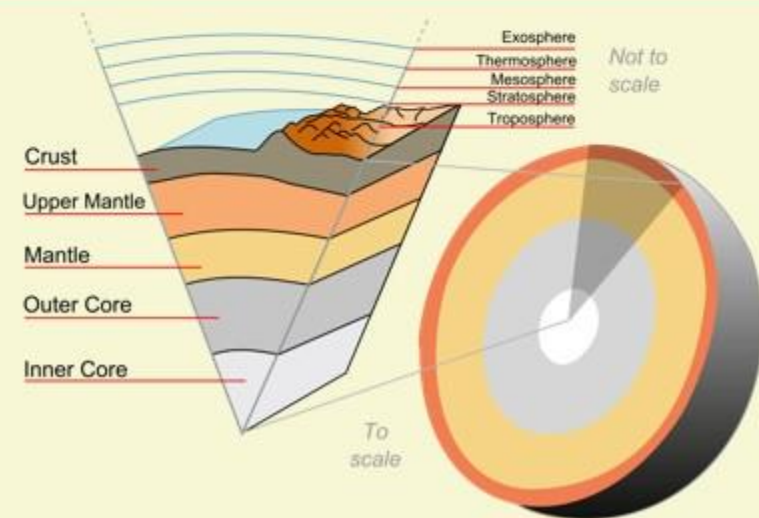
1 The outer layer of the Earth is known as Crust. It comprises two distinct parts.

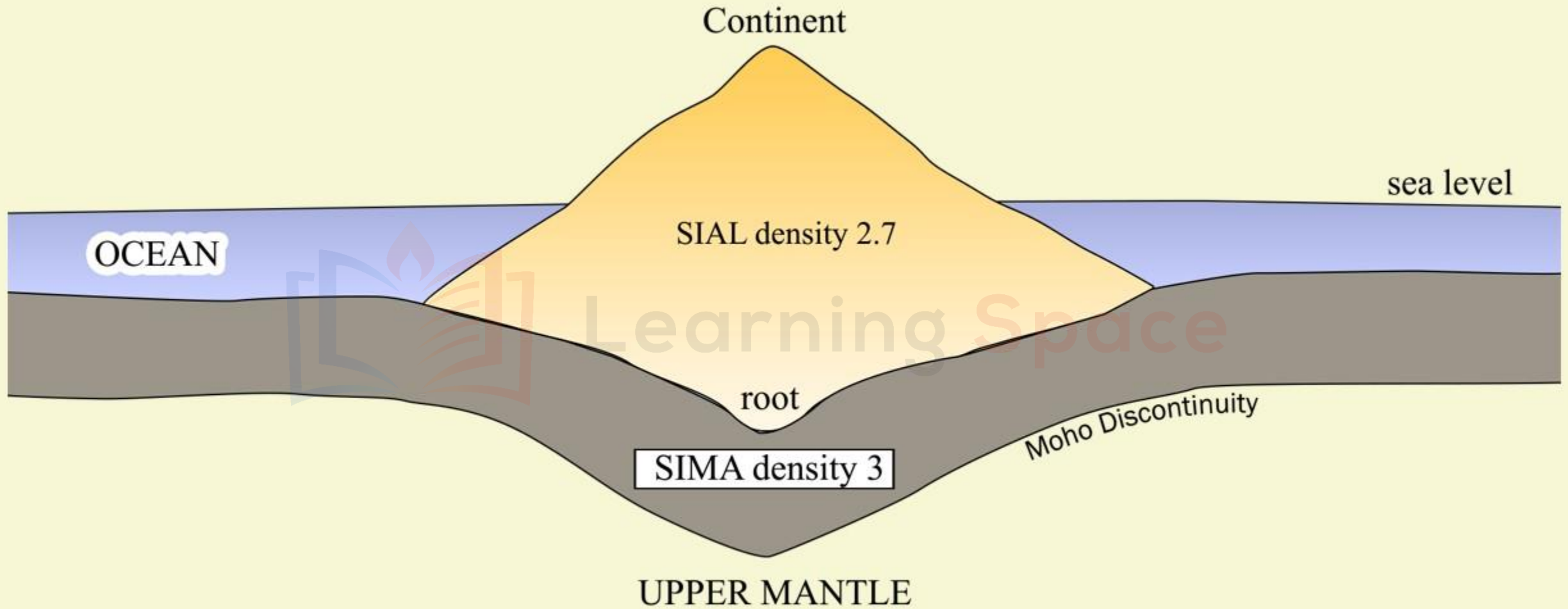
2 The upper part consists of Granitic rocks and forms the Continents. Its main mineral constituents are silica and alumina, which is referred to as “Sial”, with average density of 2.7.

3 The lower part is a continuous zone of dense basaltic rocks forming the ocean floors comprising mainly, silica, iron and magnesium. It is called “Sima”. The average density of sima is 3.

4 The Sial and Sima together form the Earth's crust, the thickness varies between 5-10 km beneath the oceans to as much as around 50 km under some parts of the Continents.

5 Sial is lighter than the Sima, hence, the Continents can be said to be “floating” on a sea of denser Sima.



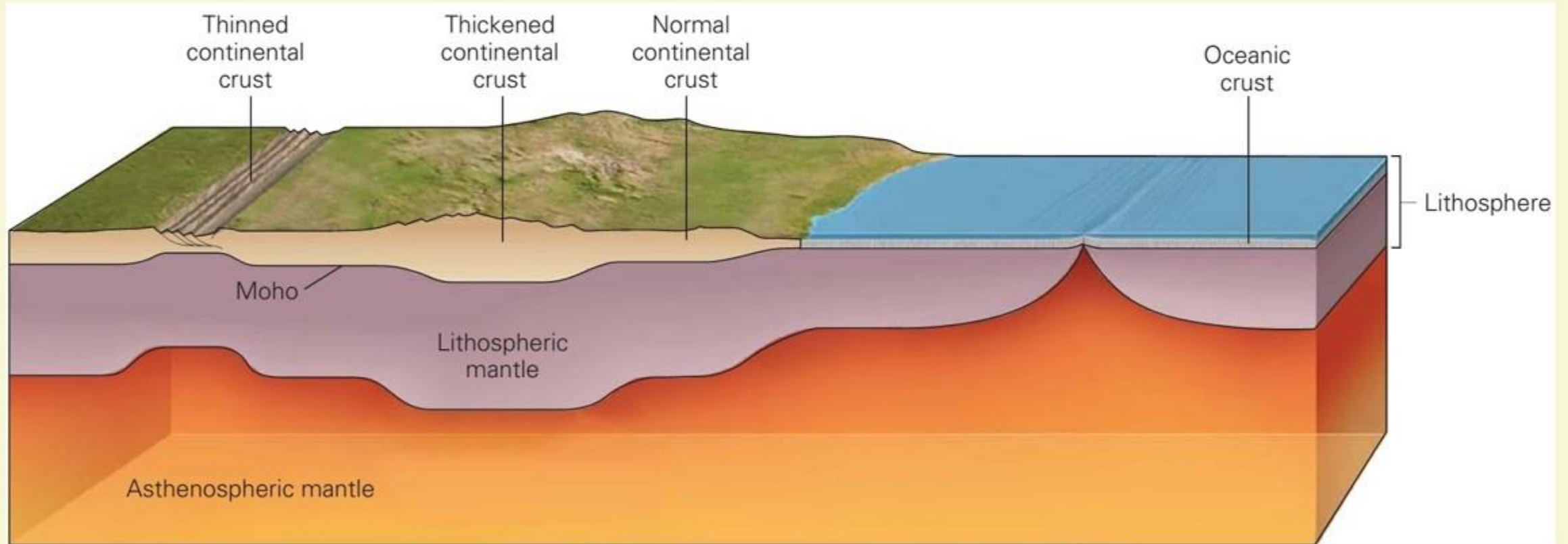


OCEANIC CRUST

This is thin layer of the Earth that exists beneath the ocean basins.

CONTINENTAL CRUST

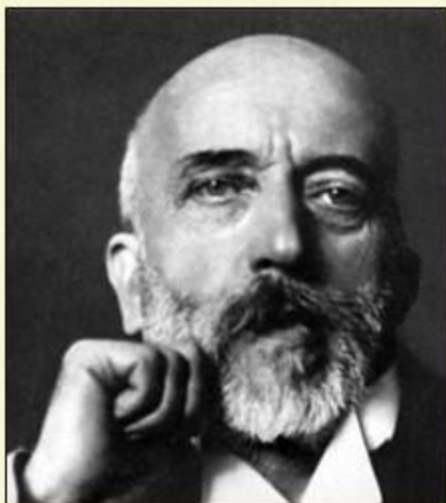
This is the thick layer of the Earth that exists beneath the Continents.



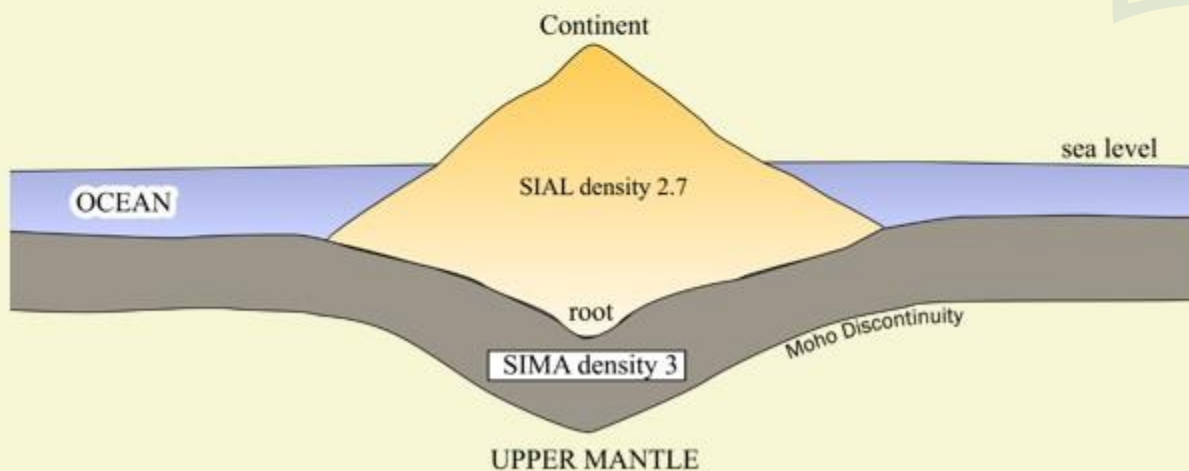


MORE ON CRUST

- 1** The Crust is made up of solid rocks and minerals.
- 2** Below the Crust is the Mantle, which is also mostly of solid rocks and minerals. But, punctuated by malleable areas of semi-solid magma.



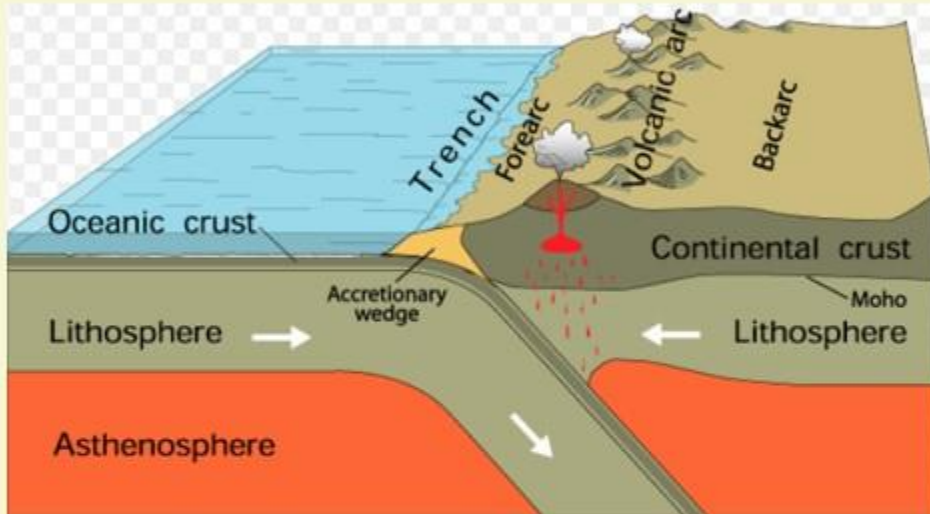
ANDRIJA MOHOROVICIC



WHAT DO YOU UNDERSTAND BY MOHOROVICIC DISCONTINUITY?

- 1 The discontinuity between the Crust and Mantle of the Earth occurring at depths, that average around 35 km beneath the Continents and about 10 km beneath the ocean floor is known as “Mohorovicic Discontinuity”.
- 2 It is named after Andrija Mohorovicic, a Croatian Geophysicist, who discovered it.
- 3 Modern instruments have determined that the velocity of the seismic waves increases rapidly at this boundary.

LITHOSPHERE ... SINGLE GEOLOGIC UNIT

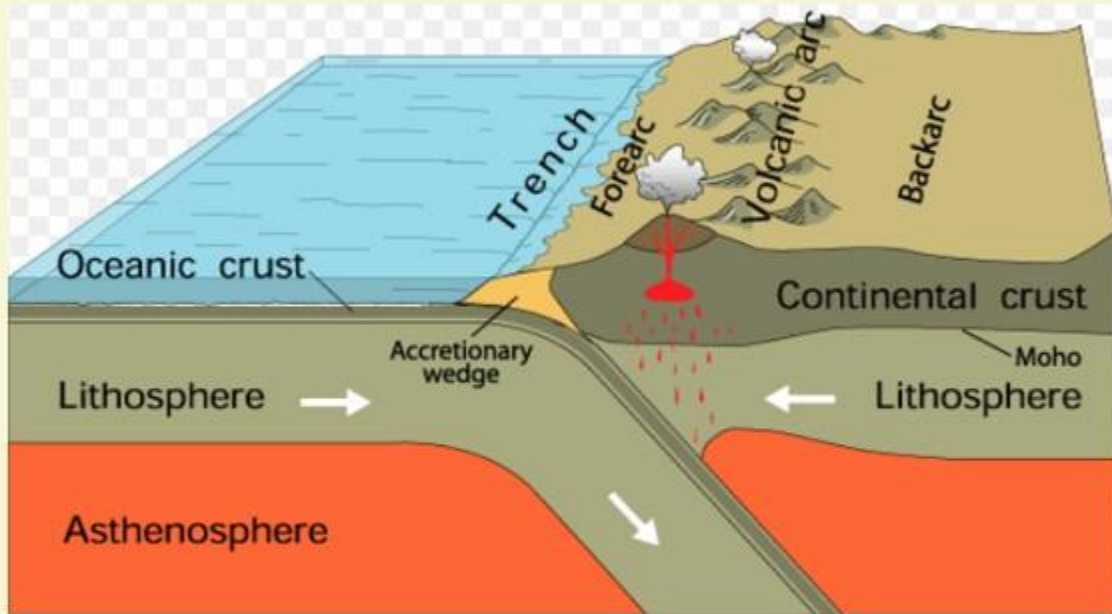


1 The crust and the upper portion of the mantle are part of the single geologic unit. This is known as Lithosphere.

2 Lithosphere's depth varies. Moreover, Mohorovicic Discontinuity i.e., the boundary between the mantle and crust does not exist at a uniform depth.

3 We can say, the Lithosphere is the solid, outer part of the Earth. It includes the brittle upper portion of the mantle and the crust. It is bounded by the atmosphere above and the asthenosphere below.

Asthenosphere, zone of Earth's mantle lying beneath the lithosphere and believed to be much hotter and more fluid than the lithosphere. It extends from about 100 km to about 700 km below Earth's surface.



MORE ABOUT LITHOSPHERE

- 1 It is the most rigid of the Earth's layers.
- 2 Although the rocks of the lithosphere are considered elastic, they are not viscous.
- 3 The underlying asthenosphere is viscous.

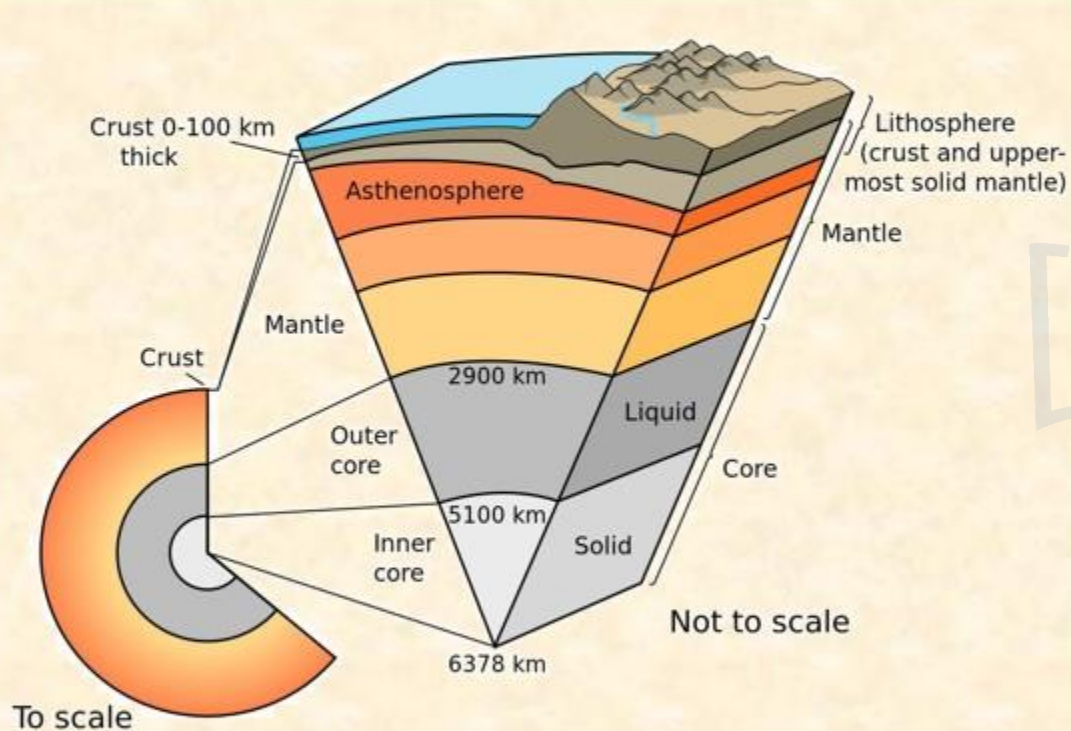
ELASTIC

An elastic material is able to stretch and be returned to its original shape or size.

VISCOUS

A viscous liquid is thick and sticky and does not flow easily.

WHAT IS THE SIGNIFICANCE OF LITHOSPHERE - ASTHENOSPHERE BOUNDARY?



1 It is the boundary, where geologists and rheologists mark the difference in ductility between the two layers of the upper mantle.

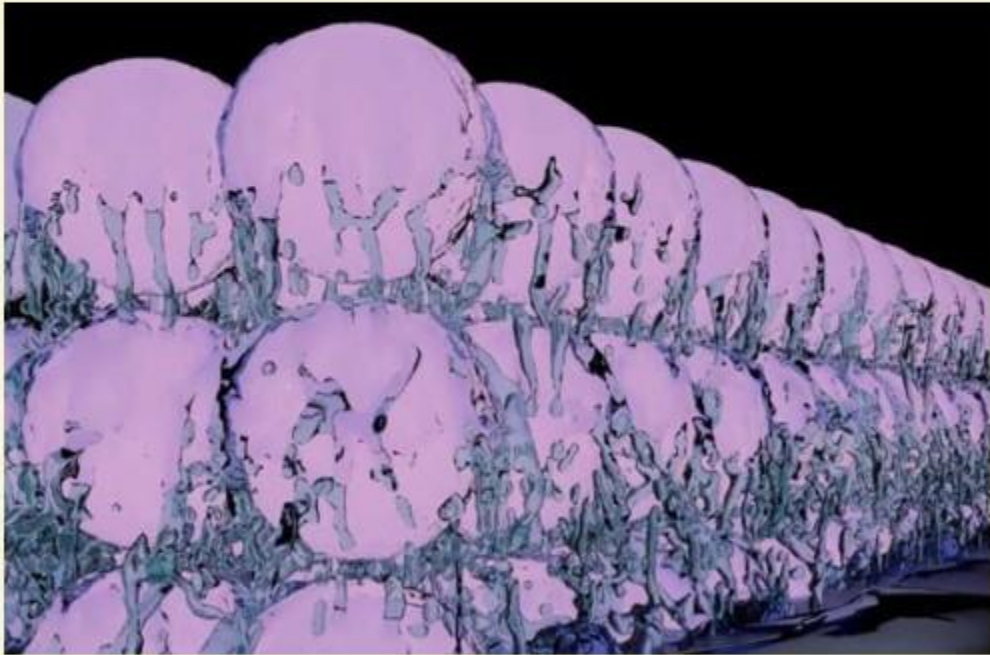
2 The lithosphere is far less ductile than the asthenosphere.

3 The elasticity and ductility of the lithosphere depends on temperature, stress and the curvature of the Earth.

The lithosphere is also the coolest of the Earth's layers.

RHEOLOGISTS

Scientists, who study the flow of matter.



DUCTILITY

It measures a solid material's ability to deform or stretch under stress.



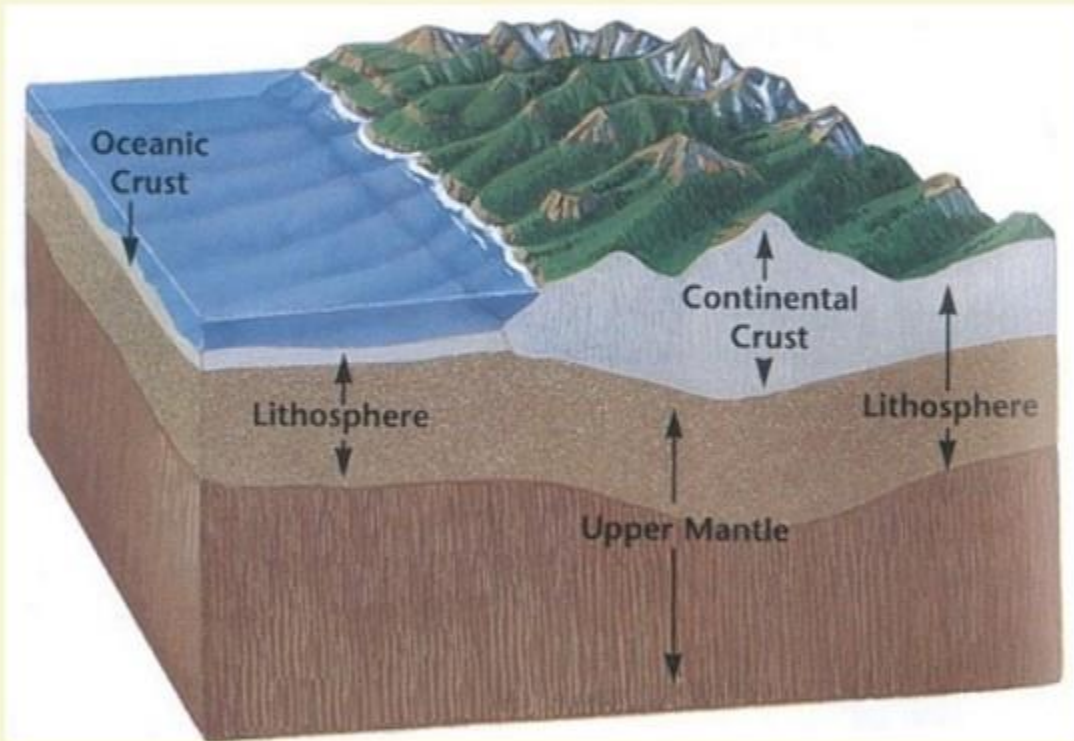
TWO TYPES OF LITHOSPHERE

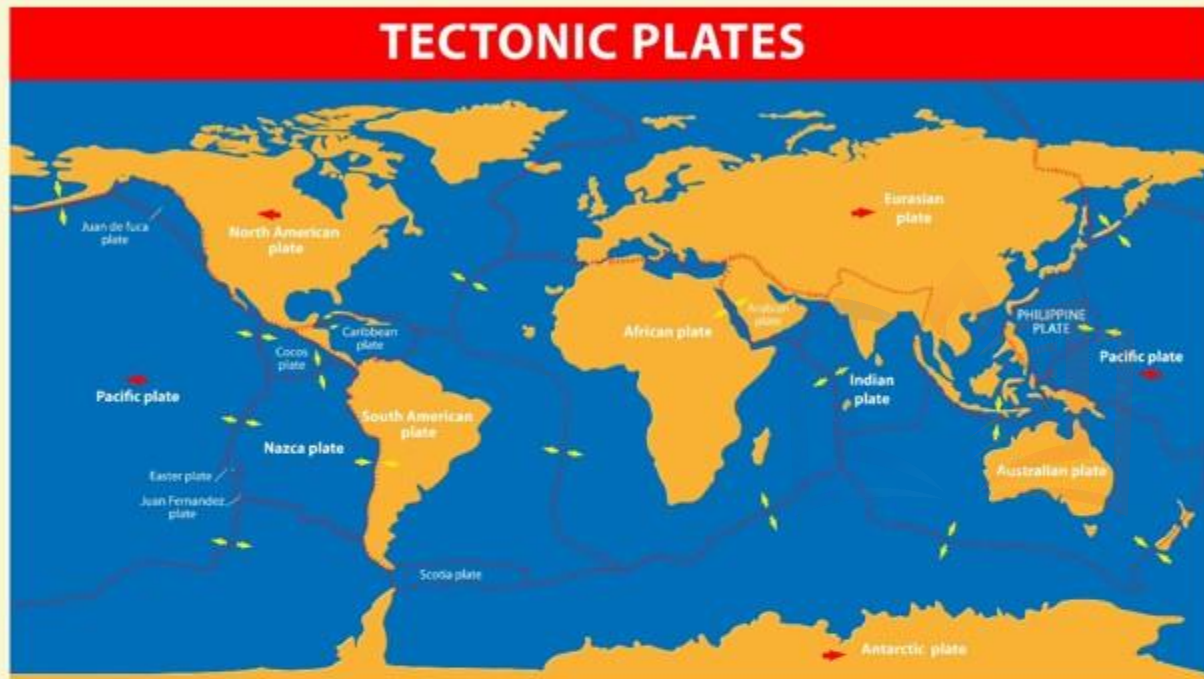
OCEANIC LITHOSPHERE

- Oceanic lithosphere is associated with oceanic crust.
- It is slightly denser than the continental lithosphere.

CONTINENTAL LITHOSPHERE

- Continental lithosphere is associated with continental crust.
- It is much, much thicker in comparison to the Oceanic Crust.
- It stretches to more than 200 km below the Earth's surface.





WHAT DO YOU UNDERSTAND BY PLATE TECTONICS?

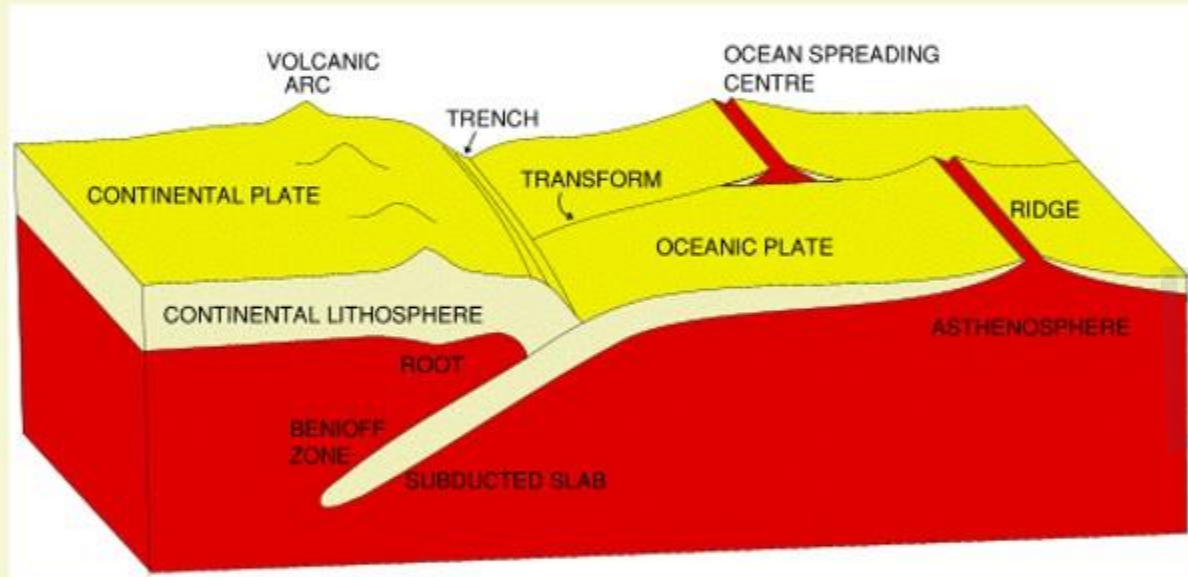
- 1 Tectonic activity describes the interaction of the huge slabs of lithosphere, called the Tectonic Plates.
- 2 Earth's lithosphere is known for tectonic activity.
- 3 Tectonic activity is the movement of tectonic plates resulting in geologic activity such as volcanic eruptions and earthquakes.

The lithosphere is divided into 15 major tectonic plates—the North American, Caribbean, South American, Scotia, Antarctic, Eurasian, Arabian, African, Indian, Philippine, Australian, Pacific, Juan de Fuca, Cocos, and Nazca.

TECTONIC PLATES



WHAT IS SO SIGNIFICANT ABOUT TECTONIC PLATES?



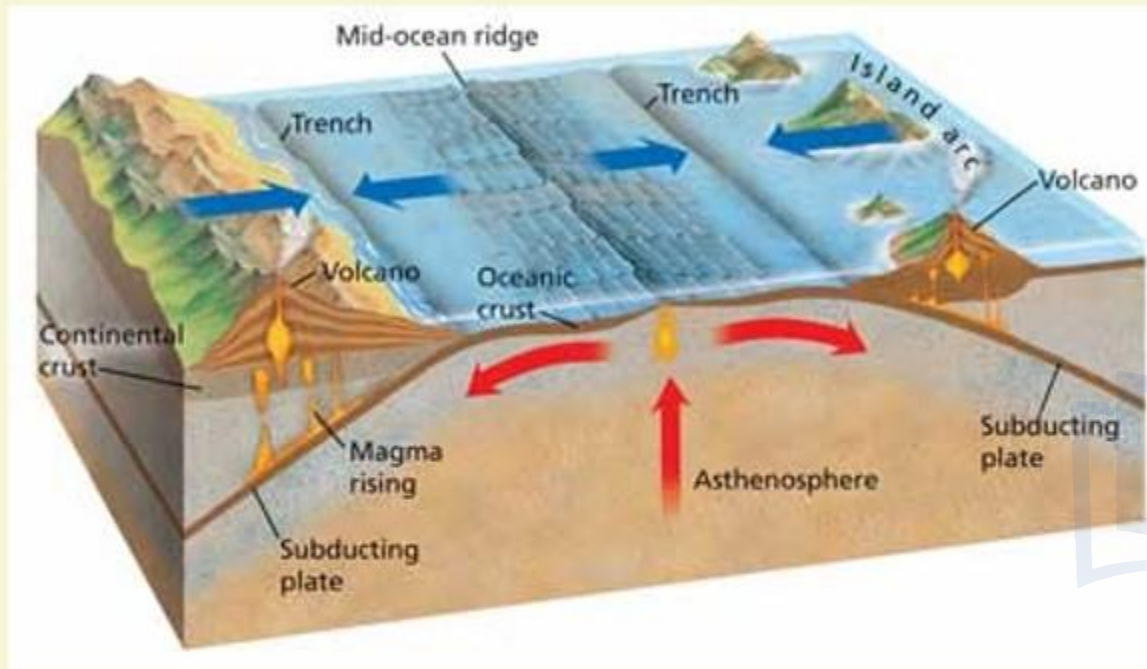
- 1** Most of the tectonic activity takes place at the boundaries of these plates. In fact, sometimes they collide or slide against each other.
- 2** The movement of tectonic plates is made possible because of thermal energy from the mantle part of the lithosphere.
- 3** Thermal energy makes the rocks of the lithosphere more elastic.



Orogeny is an event that leads to a large structural deformation of the Earth's lithosphere due to the interaction between tectonic plates.

WHAT IS THE FINAL OUTCOME OF THIS TECTONIC ACTIVITY?

- 1** The tectonic activity is mainly responsible for some of the Earth's most dramatic geologic events. We are all well aware about earthquakes, volcanoes.
- 2** Because of the tectonic activity, deep ocean trenches can be formed along with mountain-building activity.

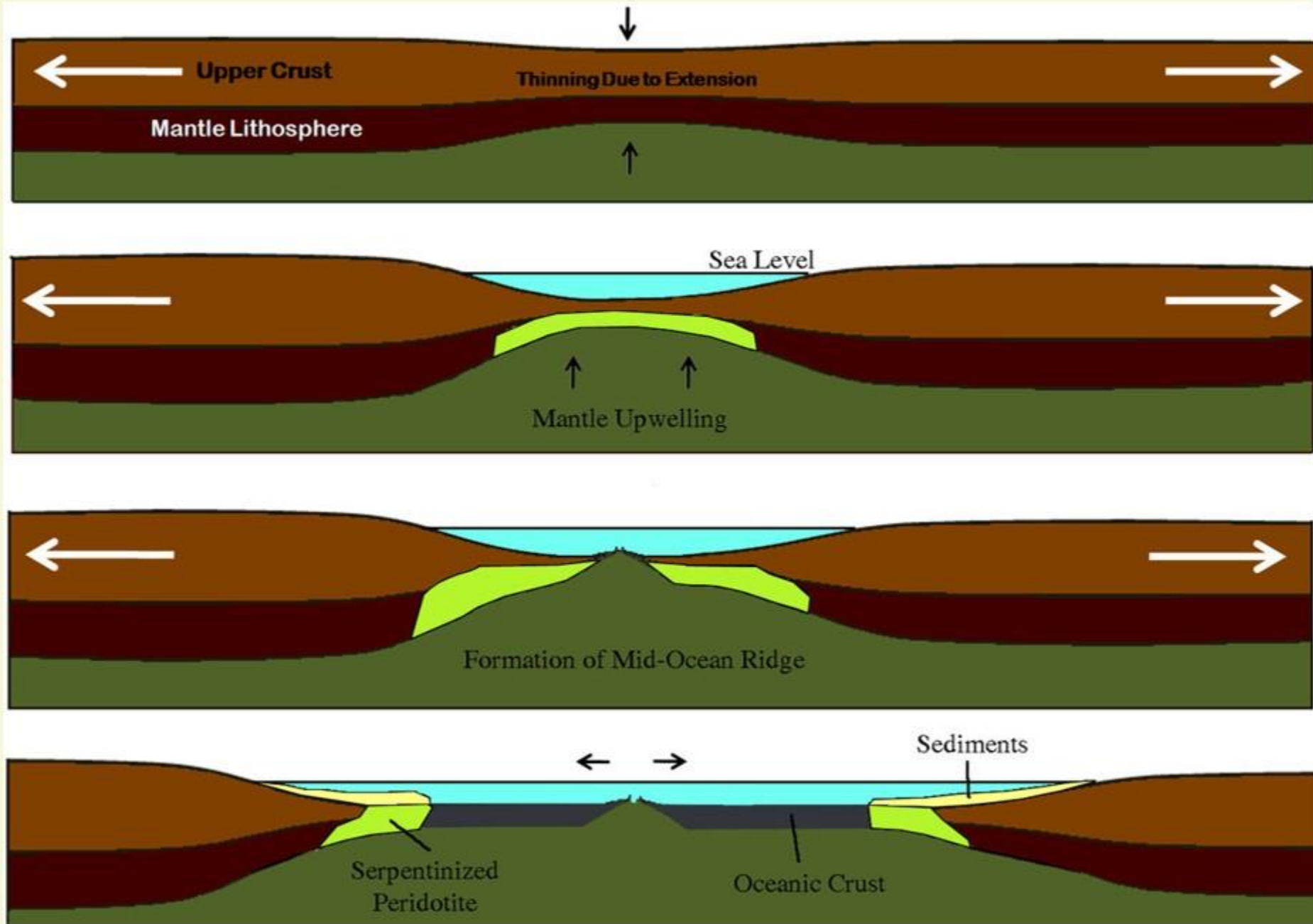


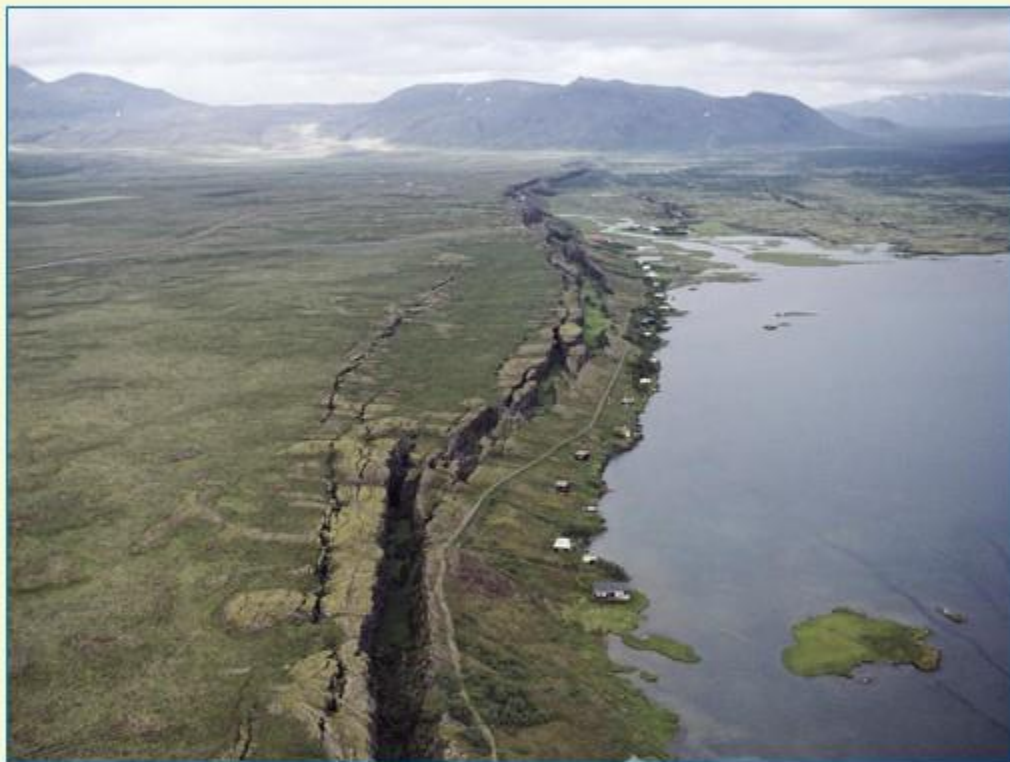
MID-OCEAN RIDGES

Underwater mountain ranges

TECTONIC ACTIVITY AND LITHOSPHERE

- 1 Tectonic activity, in fact, can shape the lithosphere itself.
- 2 Both oceanic and continental lithospheres are thinnest at rift valleys and mid-ocean ridges, where tectonic plates are shifting apart from one another.
- 3 At these zones, the lithosphere is only as thick as the crust.

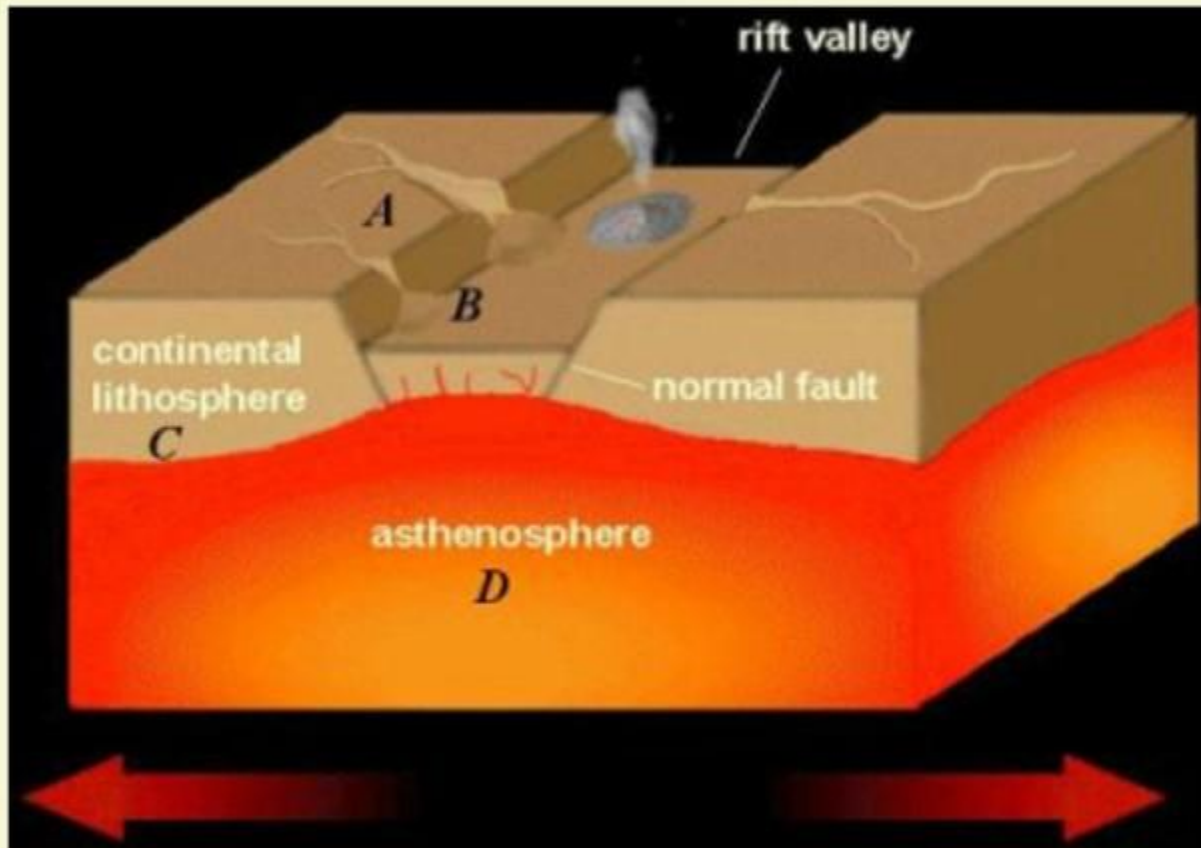




RIFT VALLEY

WHAT DO YOU UNDERSTAND BY RIFT VALLEYS?

- 1** It is the low-land region that forms, where Earth's tectonic plates move apart.
- 2** These are found both on land and at the bottom of the ocean.
- 3** Rift valleys are different from river valleys and glacial valleys, because rift valleys are created by tectonic activity and not the process of erosion.
- 4** Most rift valleys are part of “triple junctions”, a type of divergent boundary, where three tectonic plates meet at about 120° angles.



TECTONIC PLATES



SUBDUCTION

A geologic process in which one edge of one lithospheric plate is forced below the edge of another. The denser of the two plates sinks beneath the other.

MORE ON THE MOVEMENT OF TECTONIC PLATES

- Tectonic plates are huge, rocky slabs of Earth's lithosphere, its crust and upper mantle.
- Tectonic plates are constantly in motion.
 - Shifting against each other in fault zones.
 - Falling beneath one another in a process called Subduction.
 - crashing against one another at convergent plate boundaries.
 - tearing apart from each other at divergent plate boundaries.

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